```python
def filter_data(
        df: pl.DataFrame,
) -> pl.DataFrame:

    df = df.filter(
        (pl.col("defense_system_cluster_id") == pl.col("cluster_id")) &
        (pl.col("Organism") == "Homo sapiens (Human)")
    )
    logger.info(f"N rows after filtering: {len(df)}")

    return df

# What are the species anno
print("Unique values for
print(df_def_cleaned["Or

df_def_filtered = filter_
    df=df_def_cleaned,
)

df_def_filtered.write_cs
    f"{args['out_path']}.
    separator="\t",
    include_header=True
)
```

IEI project: update 2

Julian Moran

2026-04-10

The Centre for Applied Genomics, SickKids

Review – next steps

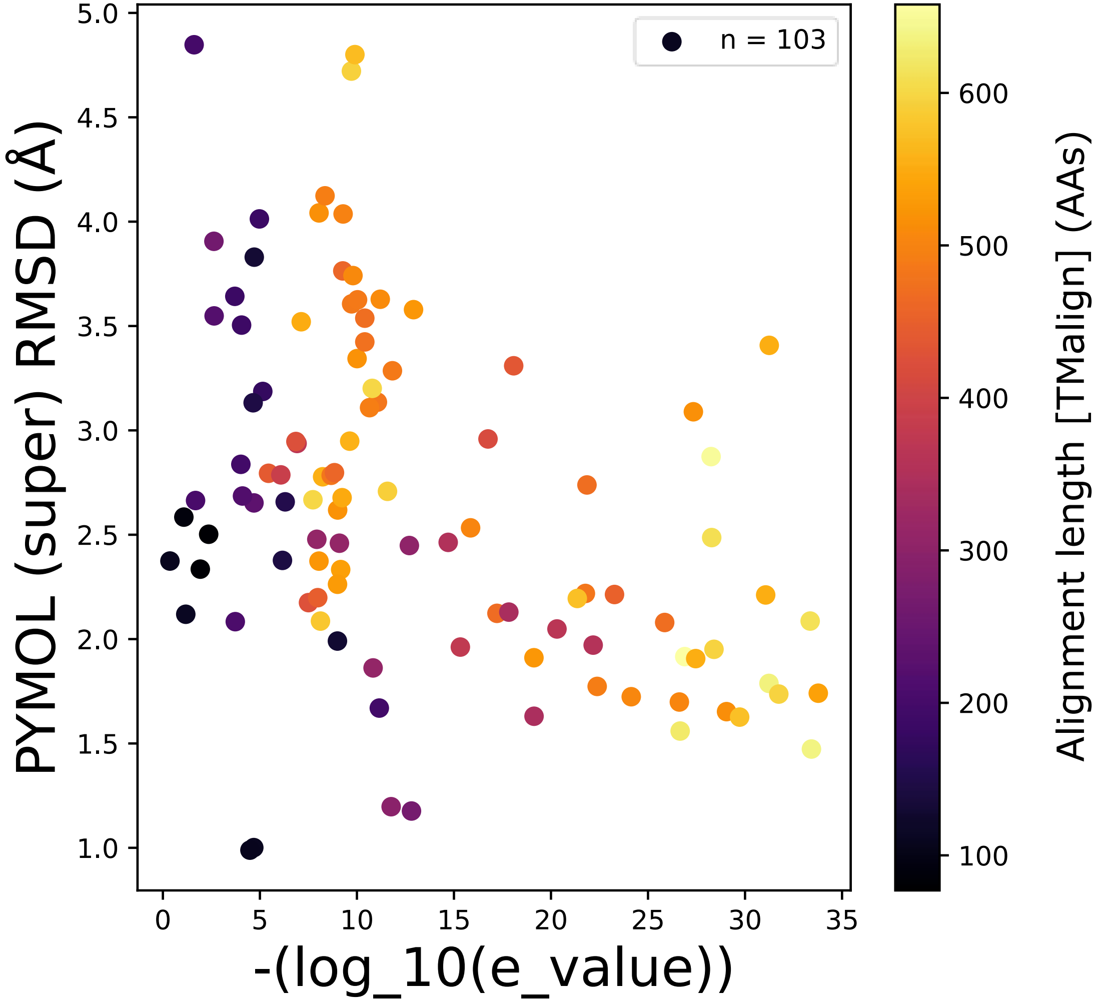
1. Automate pairwise structural alignment with PyMol

2. Finalize manually curated list of gold-standard bacteria-human protein pairs
    - source: high-impact literature
    - purpose: to serve as positive control in subsequent analysis
    - criteria: bacterial homology to human, proof of bacterial immune function

3. Build a user interface for interactive analysis
    - target users: C Trost lab, J Moran, bacterial protein research community
    - features: structural alignment visualization, sequence alignment visualization, domain annotation overlay

4. Get gene list
    - target users: C Marshall clinical research group
    - purpose: improve diagnostic yield for rare single-variant immune disease from genome sequencing

Why use PyMol to check structural alignment?

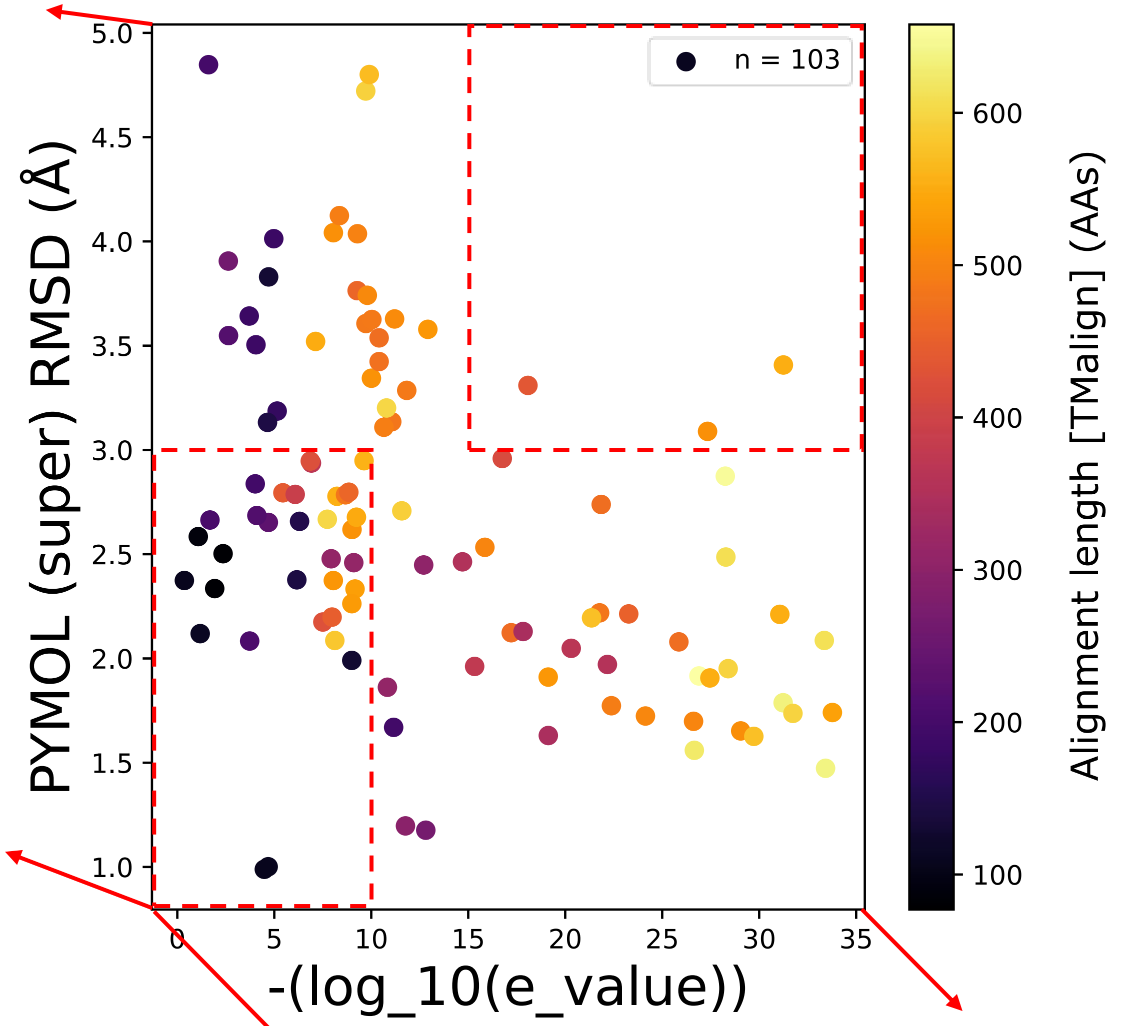


e-values [given by Foldseek] denote the probability of observing this level of structural alignment due to random chance

They are contingent on alignment length, which means filtering for only the most significant e-values risks missing domain-specific alignment

Why use PyMol to check structural alignment?

Worse structural alignment according to PyMol



Better structural alignment according to PyMol

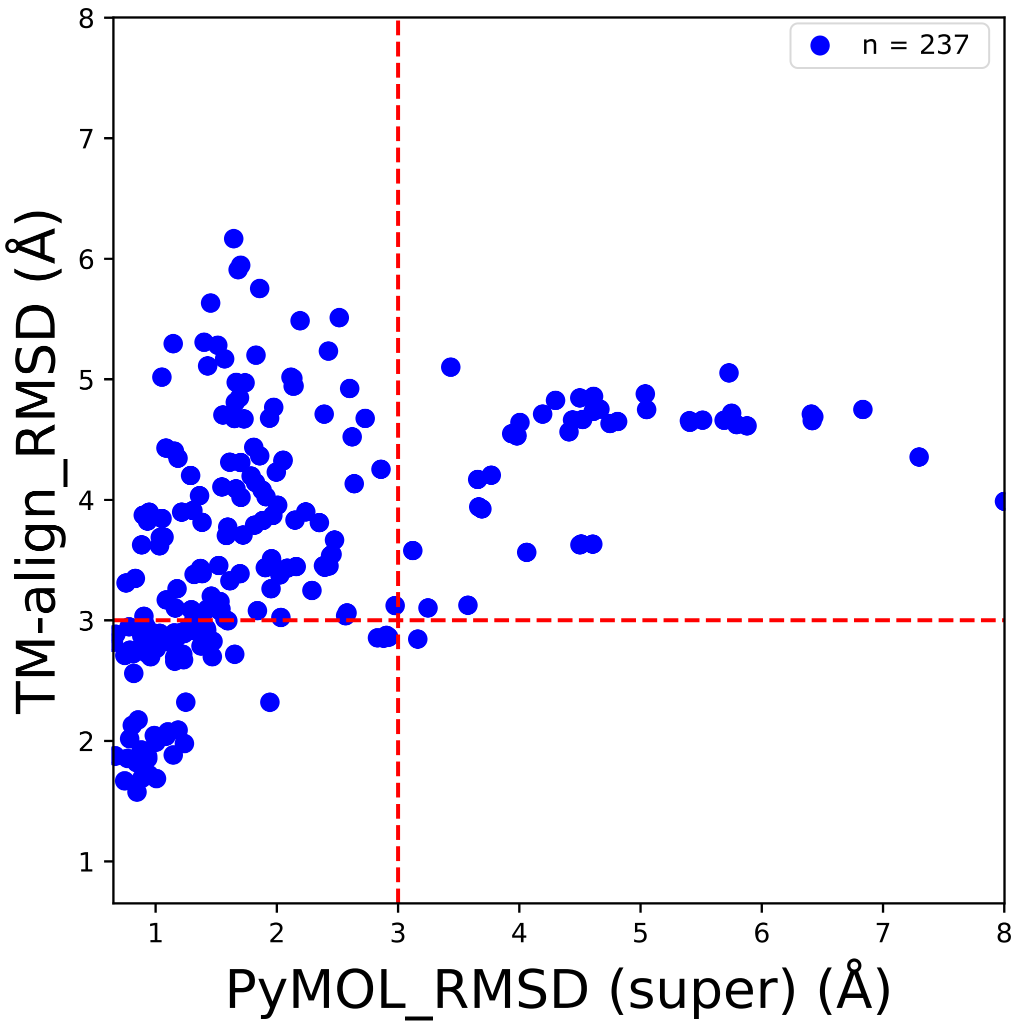
Worse structural alignment according to Foldseek

Better structural alignment according to Foldseek ($e = 10^{-35}$)

e-values [given by Foldseek] denote the probability of observing this level of structural alignment due to random chance

They are contingent on alignment length, which means filtering for only the most significant e-values risks missing domain-specific alignment

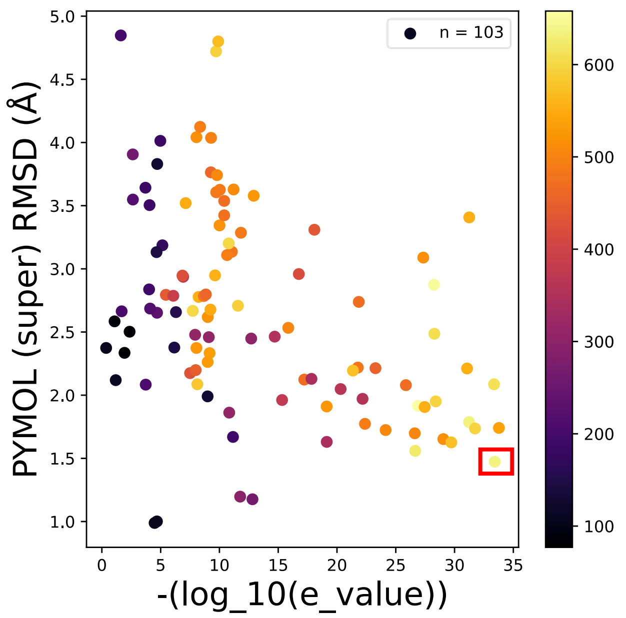
Why use PyMol to check structural alignment?



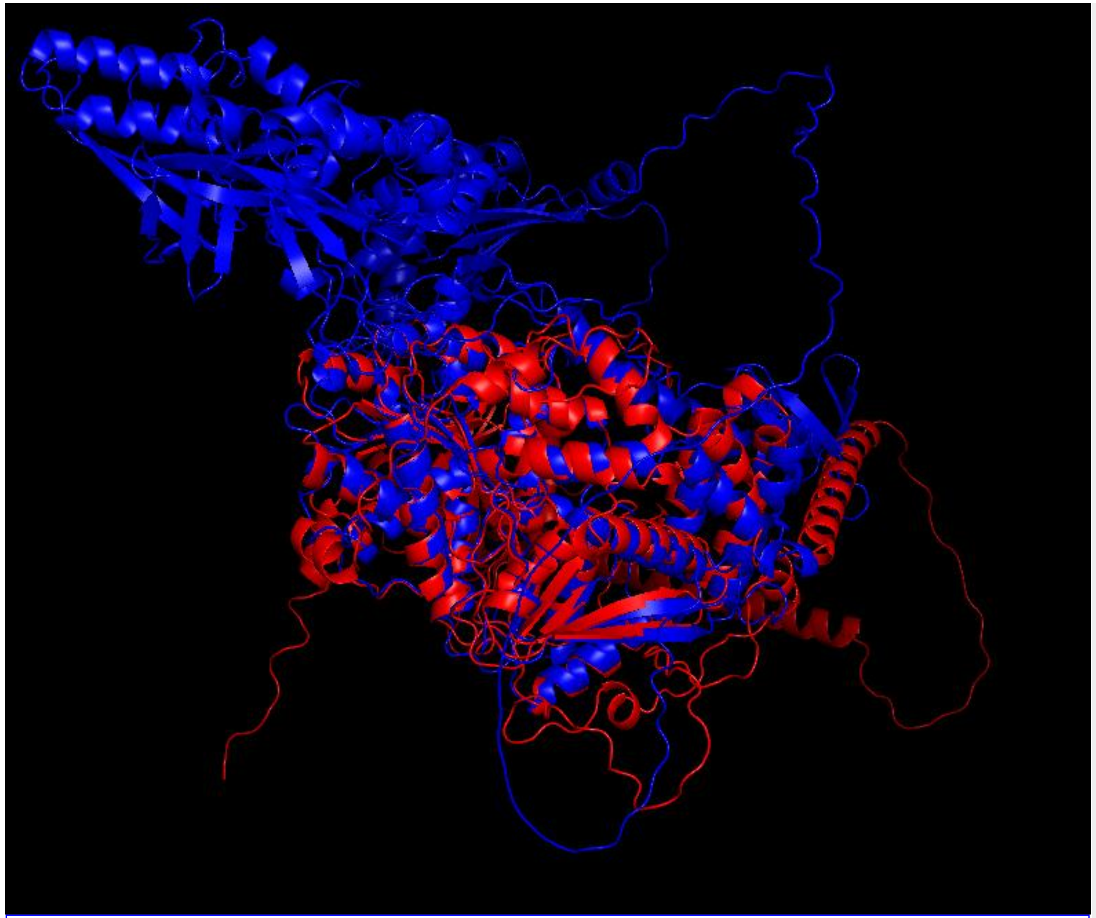
PyMol super can achieve good alignment when TM-align cannot
(see top-left quadrant vs. bottom-right).

However, this should always be checked against a threshold of $e < 0.05$ (after multiple hypothesis correction) to confirm that the alignment is not due to random chance.

To this end, we have incorporated PyMol super into our pipeline for interactive filtering



System	Zorya
e-value	$3.8e^{-34}$
PyMol RMSD	1.47 Å
Aln length	636 AAs

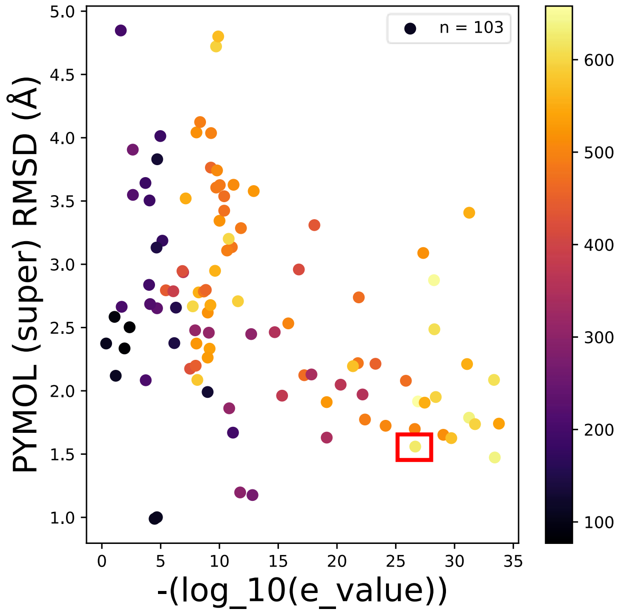


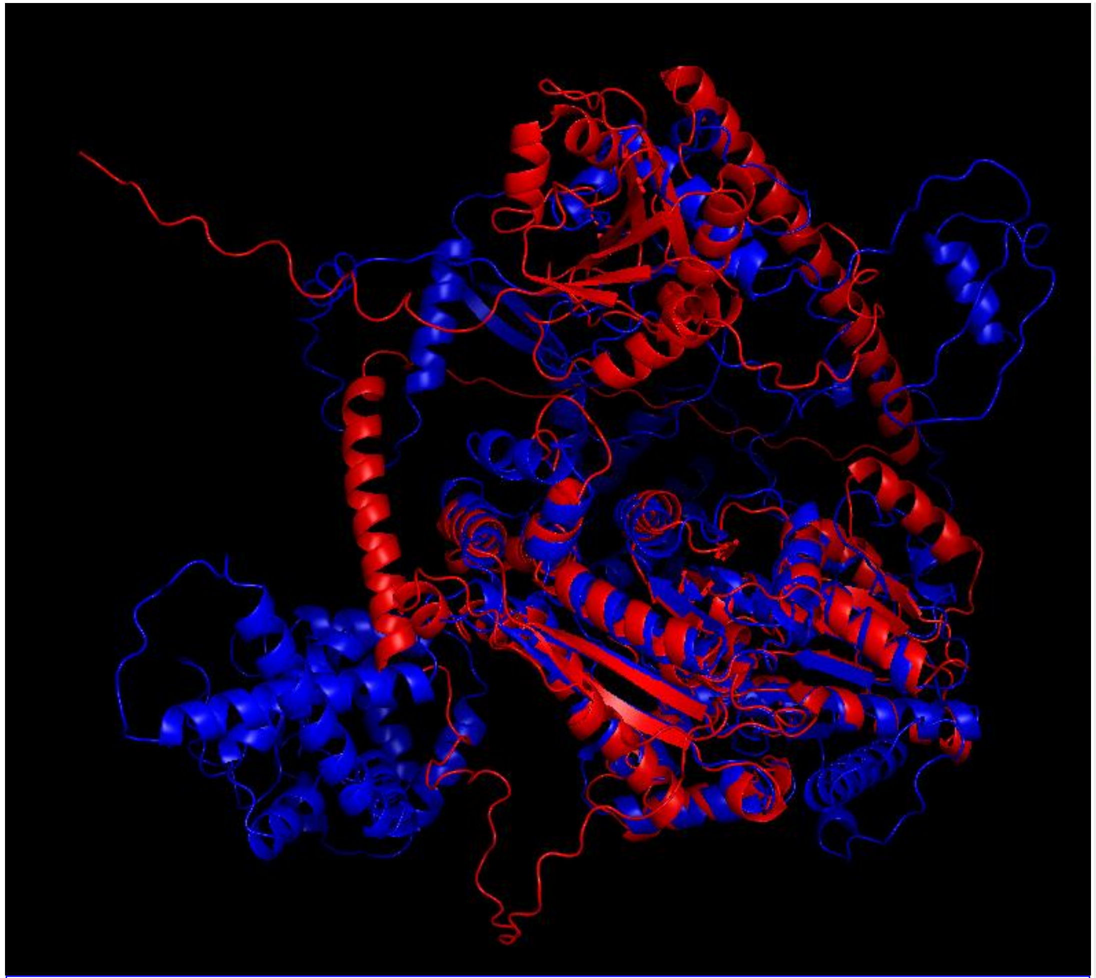
A0A0H4VDC2 [A. atlanticus]

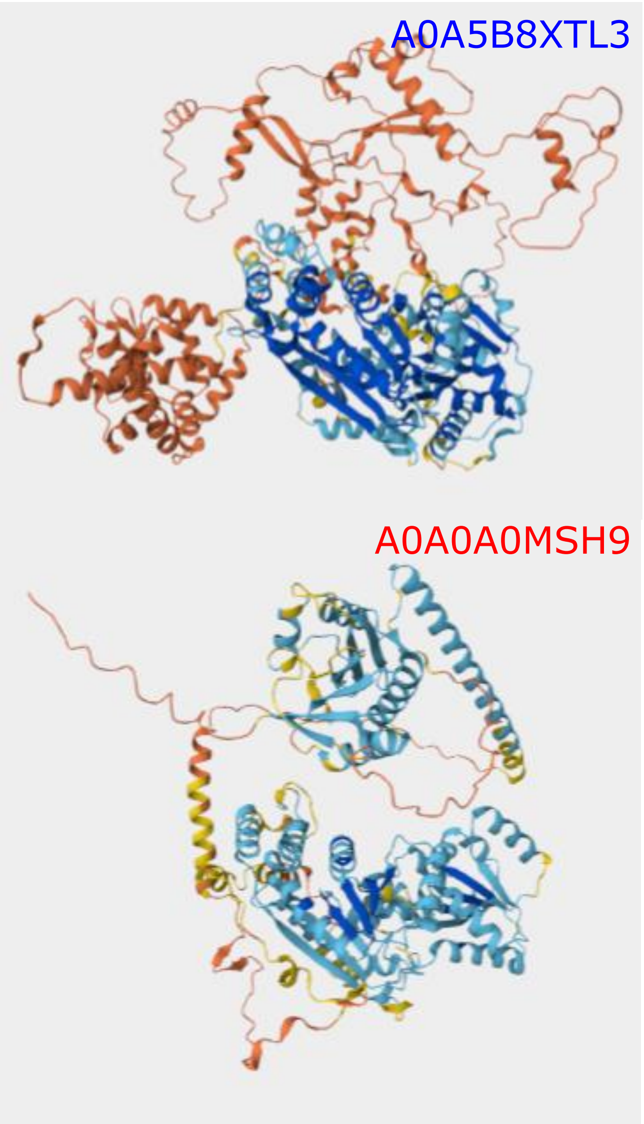
- unreviewed, uncharacterised protein
- strong sequence homology to ATP-, DEAD-box helicases in other bacterial species

X5D2E5 [H. sapiens][CHD1L]

- ATP-binding, helicase activity
- "binds histones that are poly-ADP-ribosylated on serine residues in response to DNA damage"







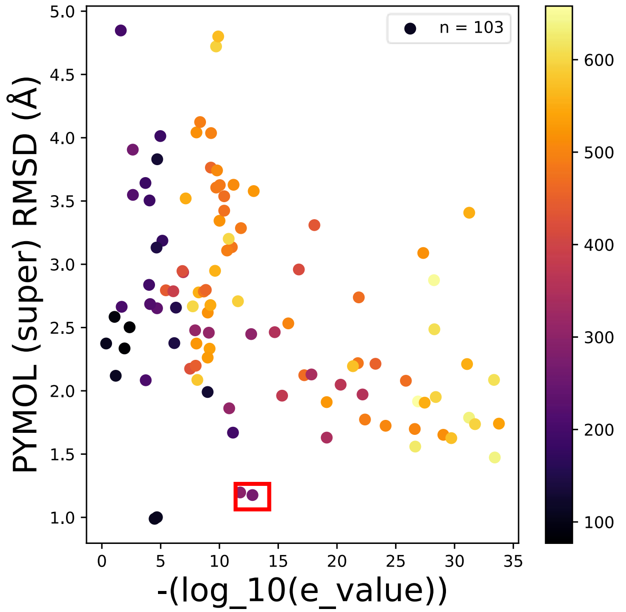
System	Zorya
e-value	$2.21e^{-27}$
PyMol RMSD	1.56 Å
Aln length	623 AAs

A0A5B8XTL3 [M. marinus]

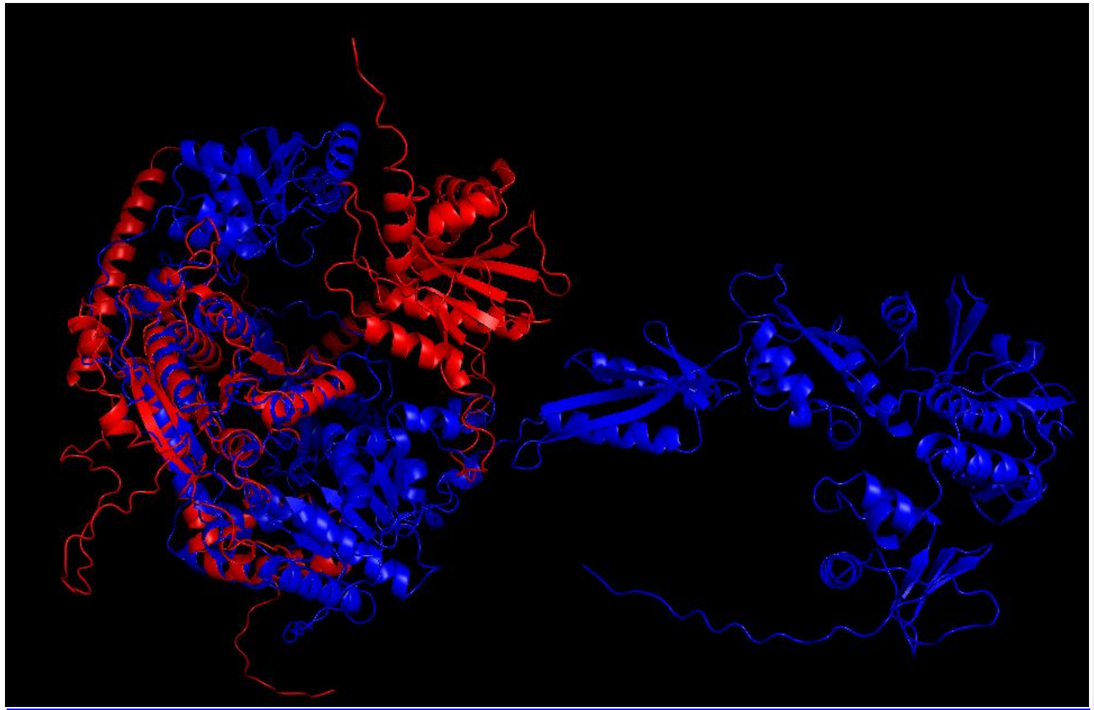
- DEAD-box helicase

A0A0A0MSH9 [H. sapiens][CHD1L]

- ATP-binding, helicase activity
- "binds histones that are poly-ADP-ribosylated on serine residues in response to DNA damage"



System	Zorya
e-value	$1.55e^{-13}$
PyMol RMSD	1.17 Å
Aln length	267 AAs

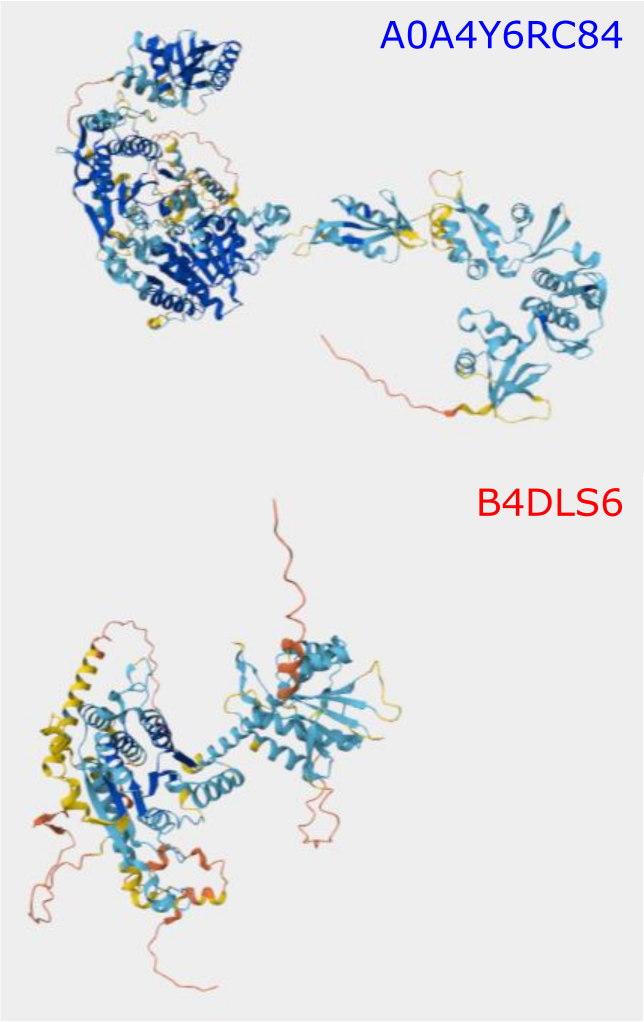


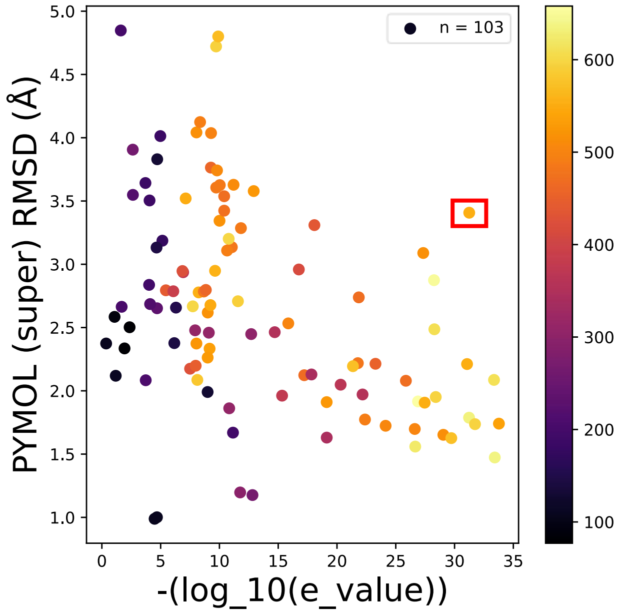
A0A4Y6RC84 [J. tructae]

- ATP-dependent helicase
- unreviewed

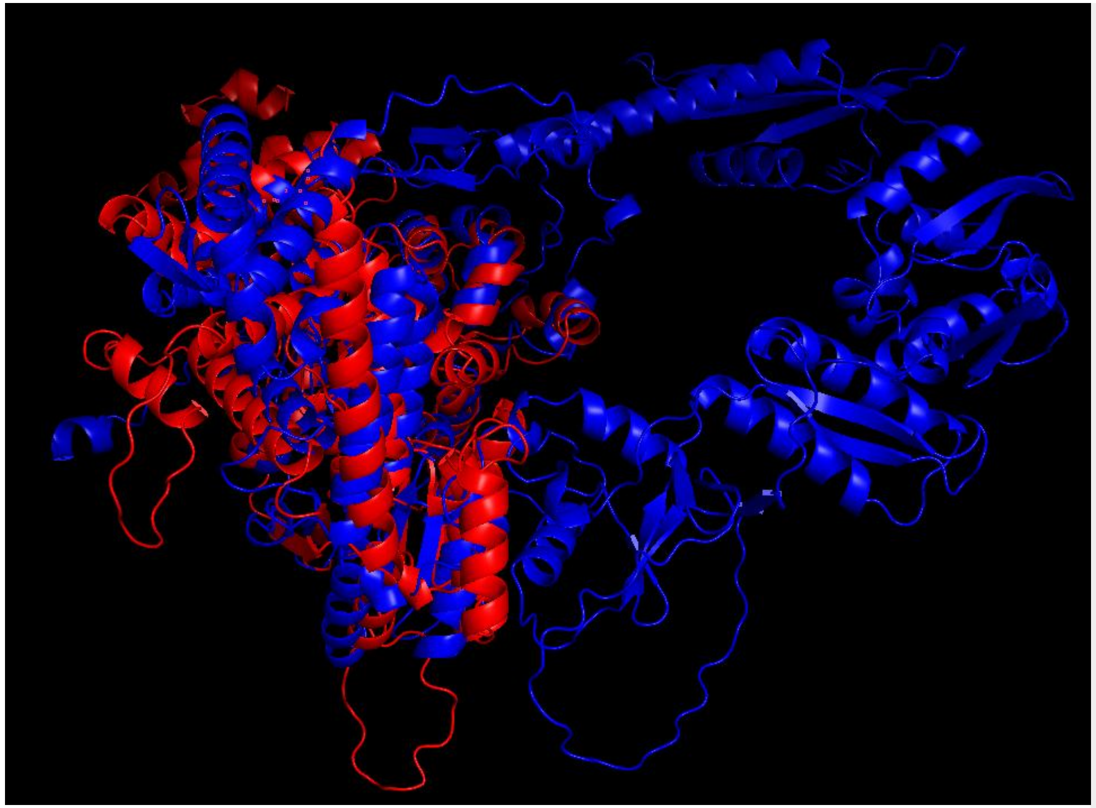
B4DLS6 [H. sapiens][unreviewed]

- "highly similar to Homo sapiens chromodomain helicase DNA binding protein 1-like (CHD1L)"



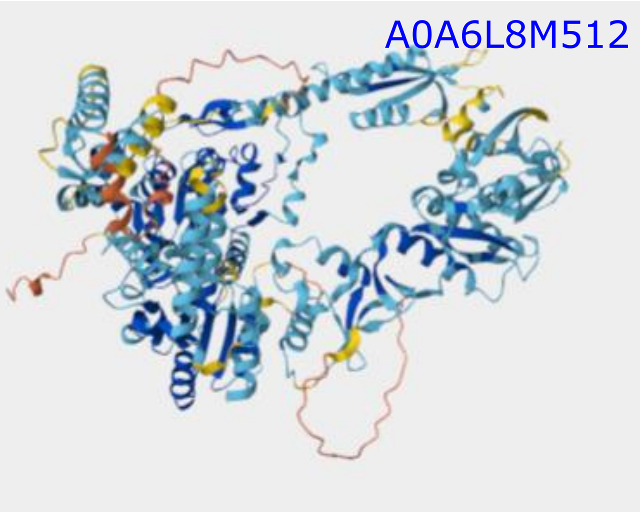


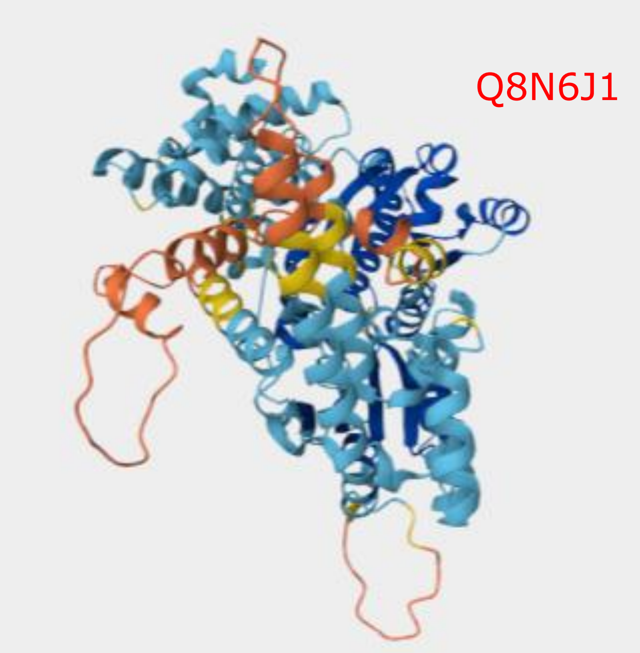
System	Zorya
e-value	$5.65e^{-32}$
PyMol RMSD	3.41 Å
Aln length	553 AAs



A0A6L8M512 [A. baumannii]
- DEAD box helicase protein

Q8N6J1 [H. sapiens][BTAF1]
- evidence of existence only at transcript level
- predicted helicase, hydrolase activity





Procuring a gold-standard list of homologous bacterial-human protein pairs from high-impact literature

Experimental criteria
provide protein accession for bacterial protein
express in new organism
perform functional assay proving bacterial immune system action

Homology criteria
provide protein accession for human protein
present convincing evidence of sequence or structure homology

Gold standard list

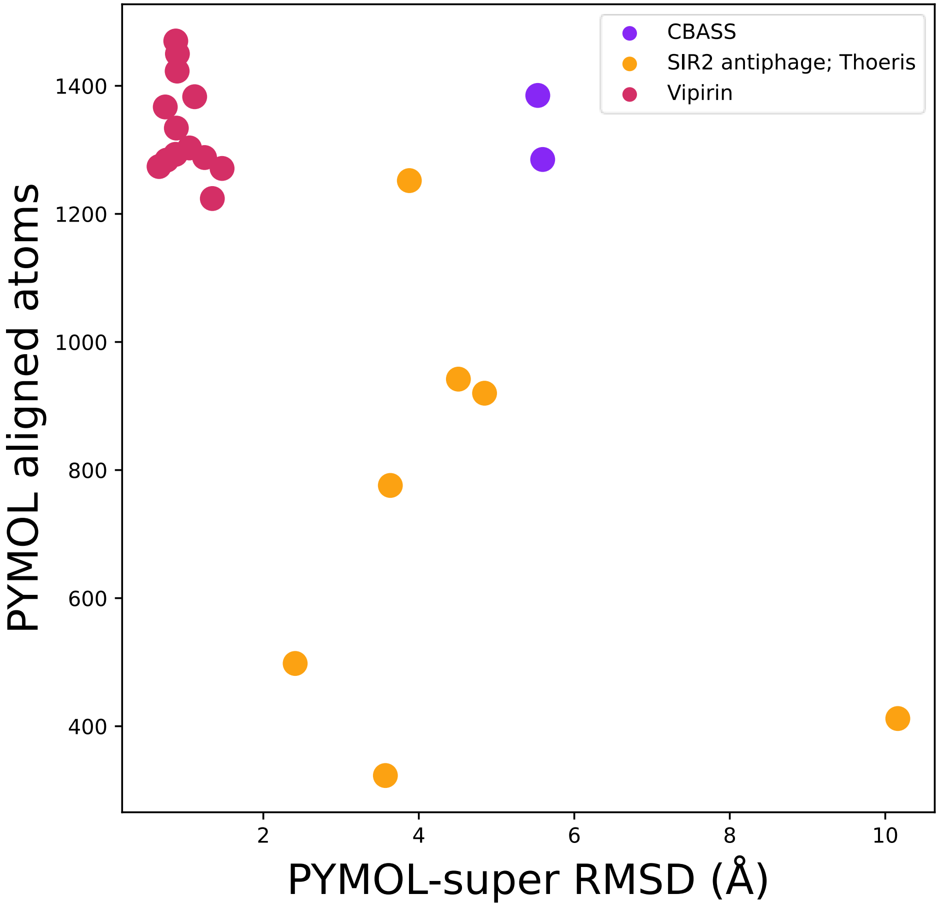
Source: high-impact literature

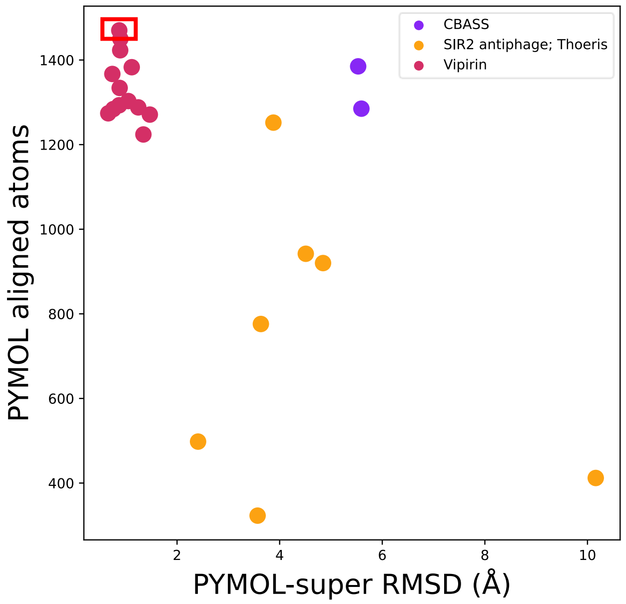
Criteria: bacterial homology to human, proof of bacterial immune function

Purpose: positive control for subsequence analysis

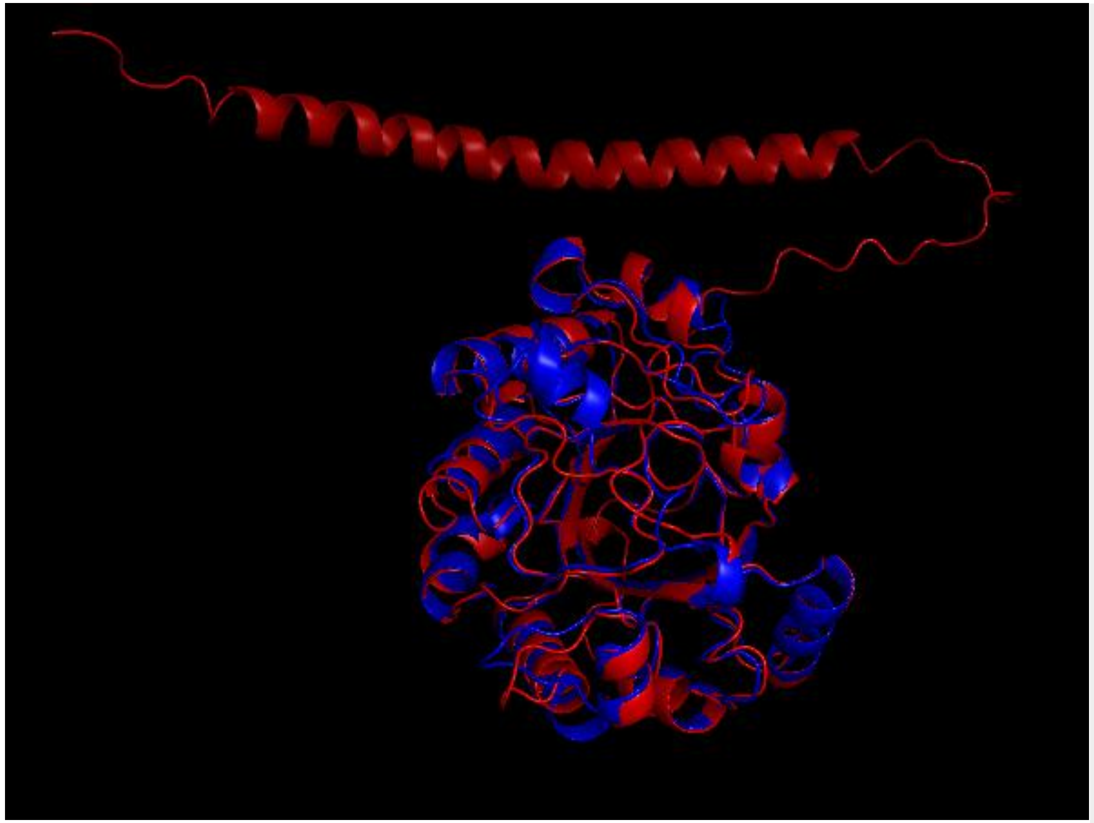
PMID	Bacterial gene name	Human gene name
30787435	cdnE	CGAS
30787435	cdnE	CGAS
40705877	ThsA	FAM118B
40705877	ThsA	SIRT1
40705877	ThsA	SIRT2
40705877	ThsA	SIRT4
40705877	ThsA	SIRT5
40705877	ThsA	SIRT6
40705877	ThsA	SIRT7
32937646	vip6	RSAD2
32937646	vip8	RSAD2
32937646	vip15	RSAD2
32937646	vip21	RSAD2
32937646	vip47	RSAD2
32937646	vip50	RSAD2
32937646	vip56	RSAD2
32937646	vip58	RSAD2
32937646	vip60	RSAD2

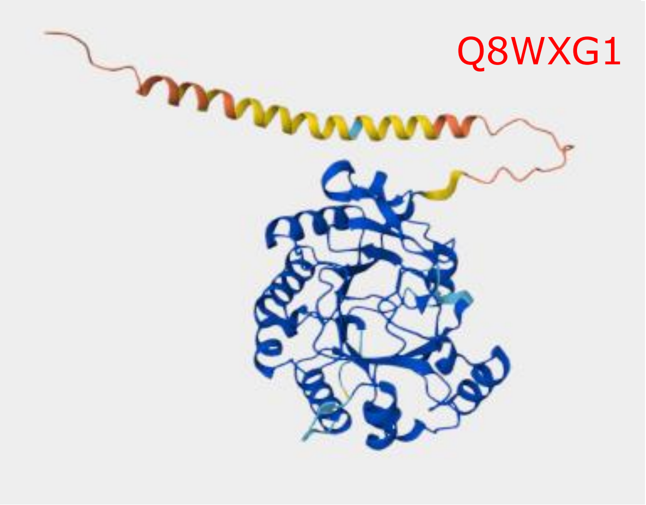
Total number: 18 pairs





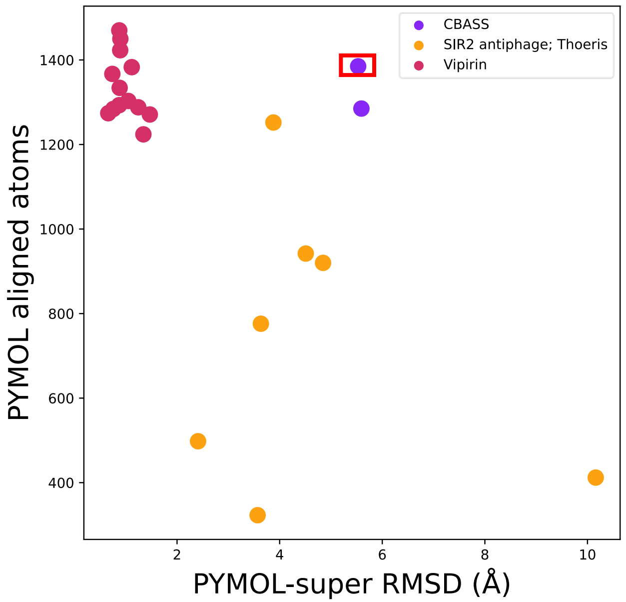
System	Vipirin
PyMol RMSD	0.89 Å
Aln length	1423 atoms [75 AAs]



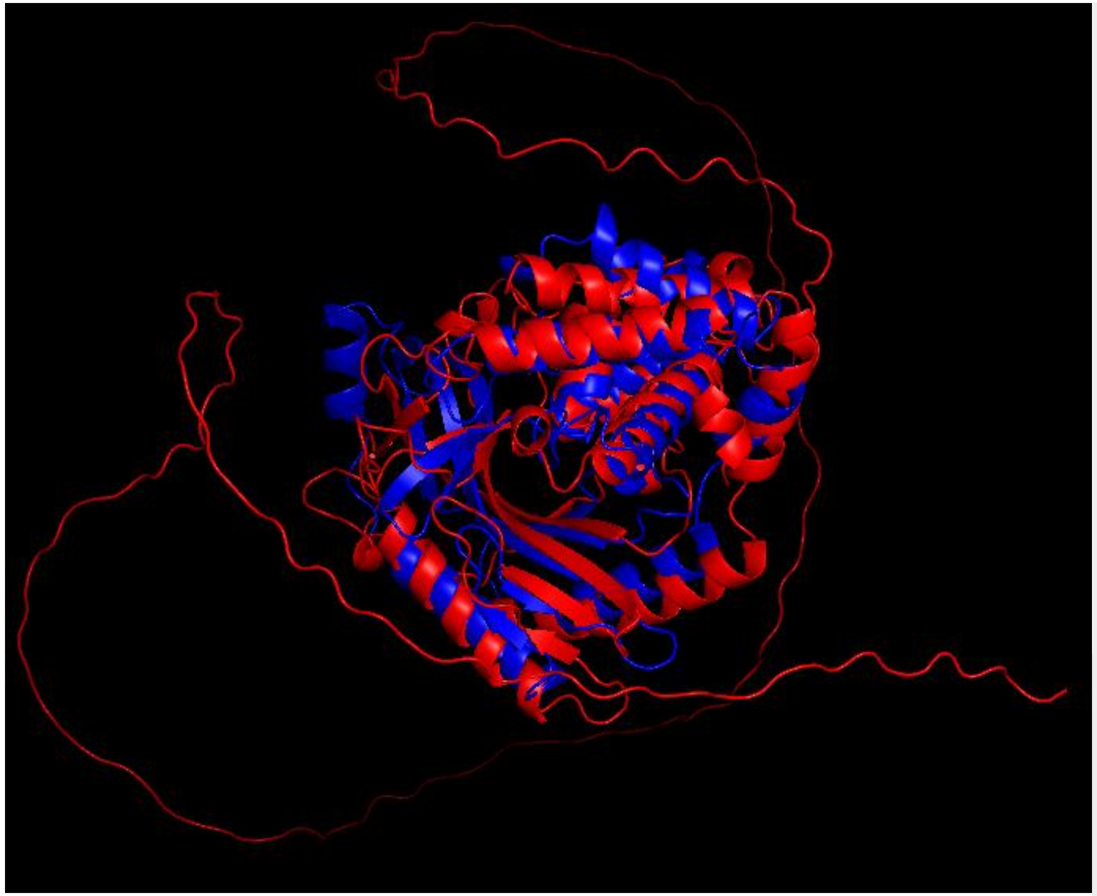


A0A1S1YUU1 [F. pacifica][vip47]
- confers resistance to phage P1
- represses T7 RNA-polymerase action

Q8WXG1 [H. sapiens][RSAD2]
- interferon-inducible antiviral protein
- initiates pathway to repress RNA polymerase action



System	CBASS
PyMol RMSD	5.53 Å
Aln length	1385 atoms [73 AAs]

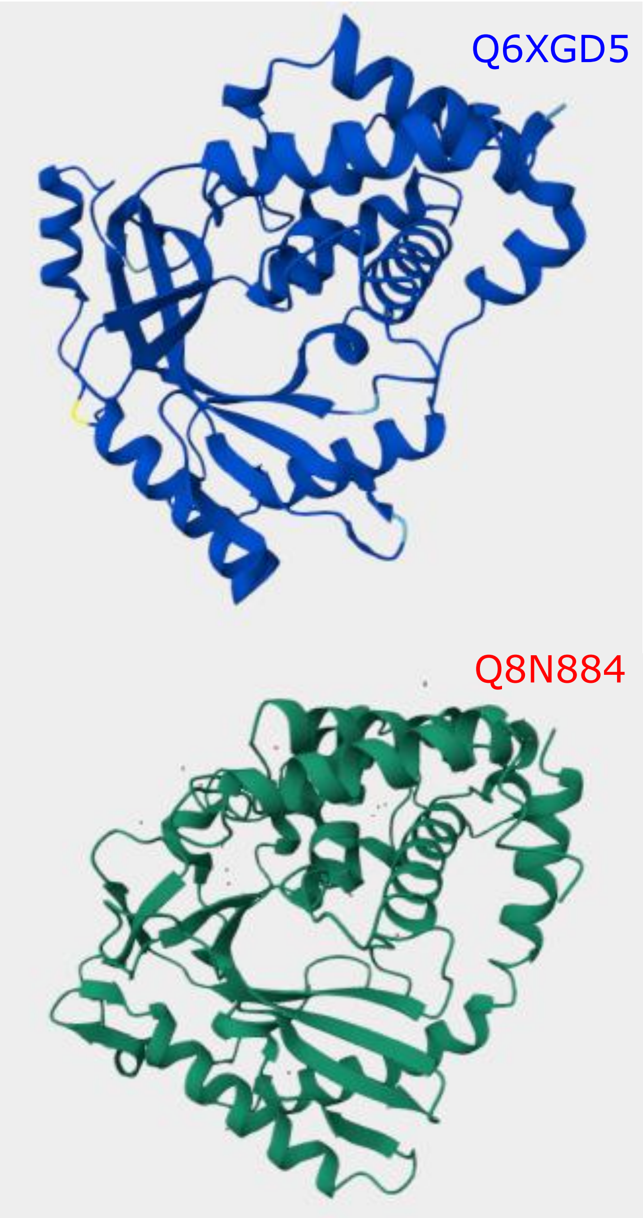


Q6XGD5 [E. coli][cdnE]

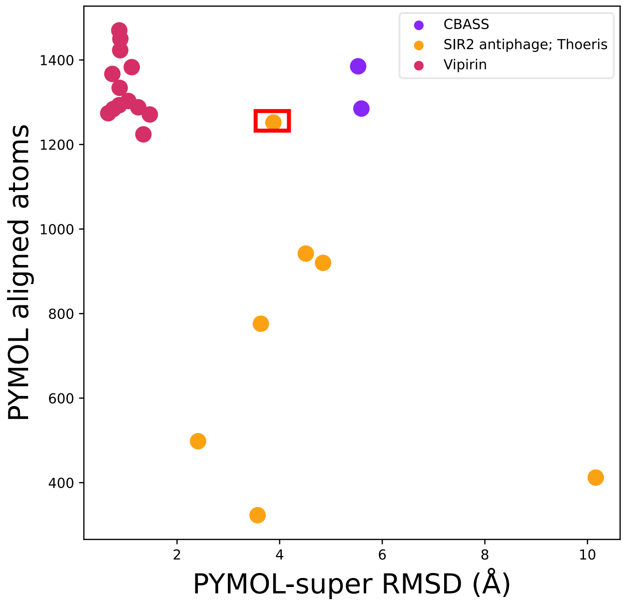
- Cyclic nucleotide synthase (second messenger synthesis) in CBASS system

Q8N884 [H. sapiens][CGAS]

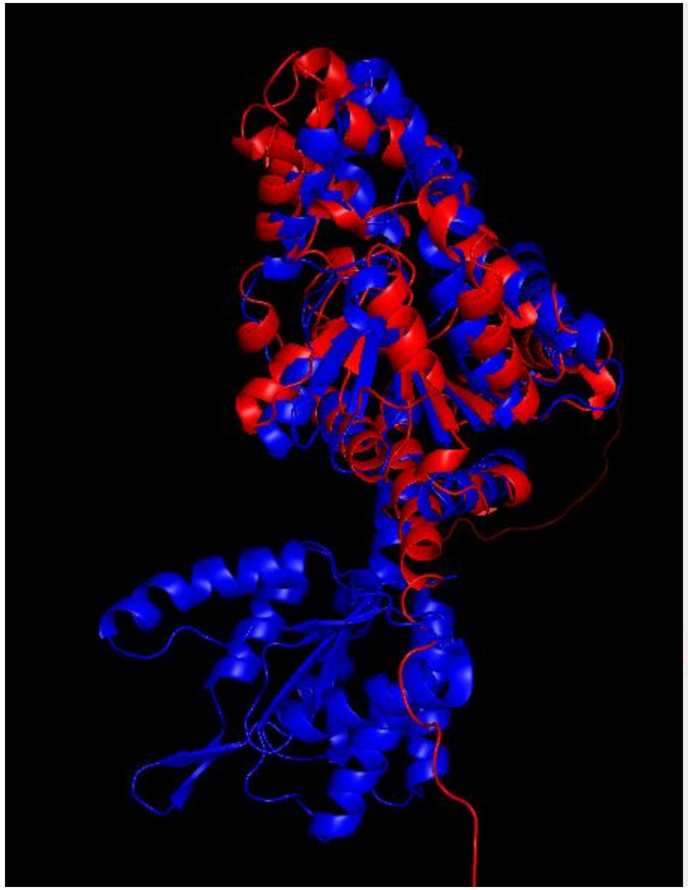
- nucleotidyltransferase catalyzing formation of cyclic GMP-AMP; plays role in innate immunity

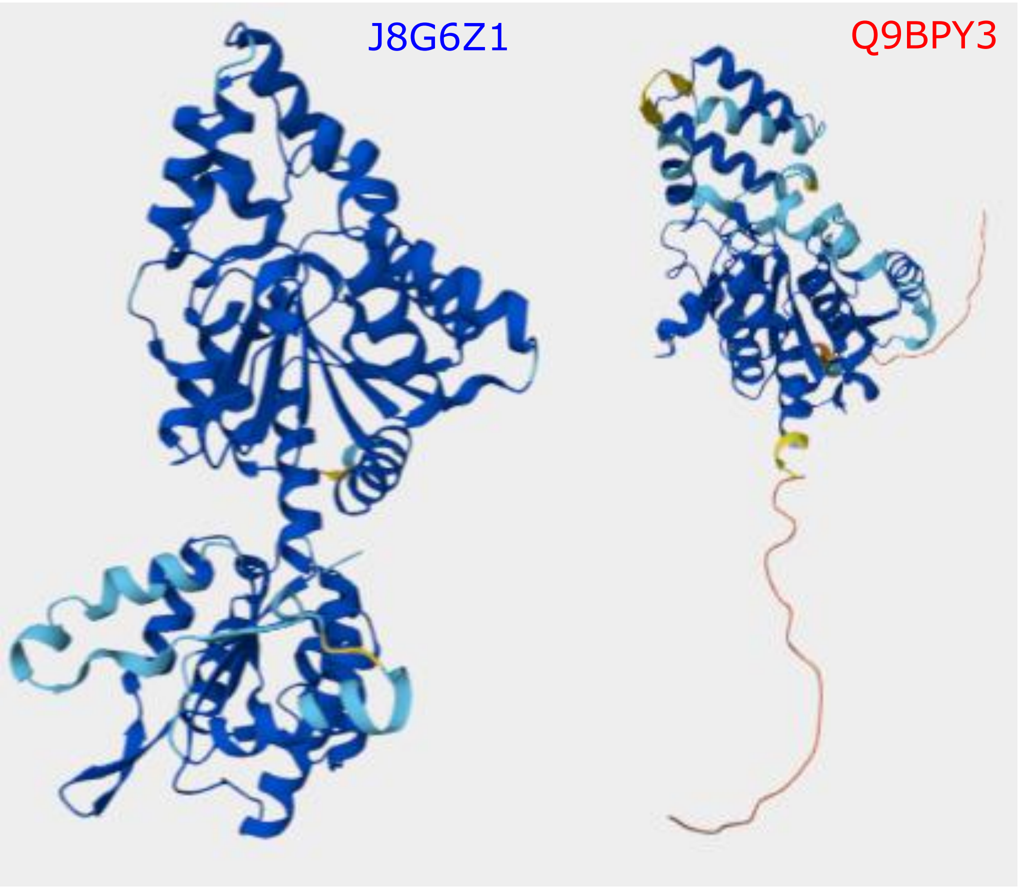


X-ray cryst. [PDB]



System	SIR2 antiphage
PyMol RMSD	3.88 Å
Aln length	1252 atoms [66 AAs]





J8G6Z1 [B. cereus][thsA]

- NAD+ hydrolyzing component (NADase) of Thoeris antiviral defense system

Q9BPY3 [H. sapiens][FAM118B]

- Cajal body regulation